

Mahi Patel

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TECHNICAL SKILLS

Languages: Java, Python, JavaScript, HTML, CSS, C, C++, Bash/Shell Scripting, Elixir, Rust, Haskell

Frameworks and Tools: React, React Native, Expo, Streamlit, Git, GitHub, Swing, XGBoost, scikit-learn, Pandas, NumPy, Plotly

Databases and Cloud: Firebase Firestore, Firebase Realtime Database, SQL, PostgreSQL, Vercel

Concepts: OOP, Data Structures and Algorithms, REST APIs, CRUD Operations, CI/CD, Unit Testing, Agile, SDLC, Feature Engineering, Monte Carlo Simulation

EXPERIENCE

Software and Product Specialist Intern

May 2026 – Present

Lyvechat

Toronto, ON

- Evaluated **WhatsApp Business API** and **multi-platform livestreaming SDKs**, producing an integration feasibility brief that outlined architecture tradeoffs for a React/Firebase stack.
- Strengthened **user retention** by identifying **5+ usability gaps** through iterative build testing, translating findings into actionable engineering tickets across bi-weekly sprint cycles.
- Identified and documented **30+ reproducible bugs**, collaborating weekly with dev teams across the UK and Poland to prioritize and unblock critical fixes.
- Translated **UX research** into **engineering-ready specifications**, closing the gap between user feedback and sprint-ready tickets across bi-weekly agile cycles.
- Surfaced **3 scalability risk areas** during roadmap discussions, leading to their inclusion in the Q3 technical backlog.

PROJECTS

F1 Race Win Predictor | *Python, XGBoost, Streamlit, Pandas, GitHub Actions*

Mar 2026

- Achieved **0.088 log-loss** on a fully held-out 2025 season by training an XGBoost classifier on **15 years of F1 data** across **6,900+ race entries**, outperforming a naive baseline by **3×**.
- Prevented data leakage in rolling performance metrics using **shift(1)** feature engineering for driver form, circuit win rate, and constructor performance.
- Generated calibrated win probabilities using **10,000 Monte Carlo simulations per race**, modelling DNFs, safety cars, and race variability.
- Automated data ingestion, retraining, and deployment with a GitHub Actions **CI/CD pipeline** that redeploys predictions to Vercel after each Grand Prix.
- Built a qualifying-input prediction system for the live 2026 season, enabling real-time race win probability forecasts.

Supplement Verification & Tracking App | *React Native, Firebase*

Sep 2025 – Dec 2025

- Built a **full-stack mobile app** that analyzes supplement safety in under **3 seconds** using real-time barcode scanning and ingredient classification.
- Integrated the Open Food Facts API (**1M+ products**) with a custom scoring engine that evaluates allergens, caffeine content, and risky additives using a dynamic 0–10 safety rating system.
- Developed **multi-format barcode scanning** support for UPC, EAN, Code128, and QR formats using the **Expo Camera API** with low-light flash controls for improved scan reliability.
- Implemented **Firebase Authentication** and **Firestore synchronization** to support persistent login, guest mode, scan history, favourites, and cross-device real-time data persistence.
- Designed and deployed a personalized health dashboard with searchable scan history, usage analytics, daily supplement tips, and user-specific alerts.

EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Science in Computer Science, Co-operative Education

Sep. 2024 – May 2029

- Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Discrete Mathematics, Computer Organization and Architecture, Probability and Statistics